

HEProAI Activity 4: Detailed Process Flow

From Machine Learning Insights to Human Intervention

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1. Introduction

The objective of Activity 4 is to operationalize the machine learning insights generated in Activity 3. This phase focuses on the "Action" component of the HEPro AI+ system, ensuring that the student segmentation results in high-quality, targeted human support.

2. Phase 1: Mentor Dataset Engineering

The first step involves the creation of a specialized `mentors.csv` dataset. Each mentor profile is designed with specific metadata to facilitate accurate matching:

- **Expertise:** Categorized into Academic (APS), Wellness (WWS), and Career (CRS).
- **Capacity & Load:** Quantitative constraints to ensure no mentor is overburdened, maintaining the quality of intervention.

3. Phase 2: Intelligent Matching Algorithm

The core of Activity 4 is the matching engine. It operates on a two-step logic:

1. **Need Detection:** The system identifies a student's cluster profile. For instance, Cluster 1 (High Stress) is automatically assigned a "Wellness" requirement, while Cluster 2 (Global Risk) is assigned "Academic" support.
2. **Load-Balanced Assignment:** The algorithm filters for mentors with the required expertise and selects the one with the highest remaining capacity using the following fitness function:

$$Fitness = \max(\text{Mentor Capacity} - \text{Current Load}) \quad (1)$$

4. Phase 3: High-Risk Alert Simulation

To ensure timely response, Activity 4 implements an automated alert system. Students belonging to Cluster 1 (Burnout) and Cluster 2 (At-Risk) are prioritized.

- **Critical Alerts ():** Automatically triggered for Cluster 1 to address low Wellness scores immediately.
- **Intervention Playbooks:** Each match is accompanied by a specific strategy, such as "Workload Reduction" for stress or "Remedial Tutoring" for academic struggles.

5. The Final Output: Recommendation Pipeline

The final result of the Activity 4 flow is the **Recommendation Table**. By automating this mapping, the platform ensures that no student is left without a personalized roadmap to success.